

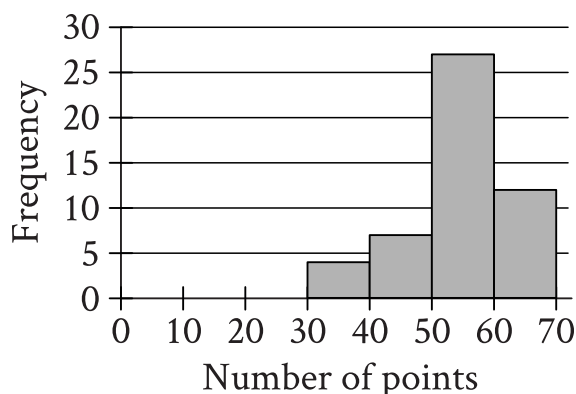
● HARD

● PROBLEM-SOLVING AND DATA ANALYSIS

One-variable data: Distributions and measures of center and spread

02

10 questions · ~15 min

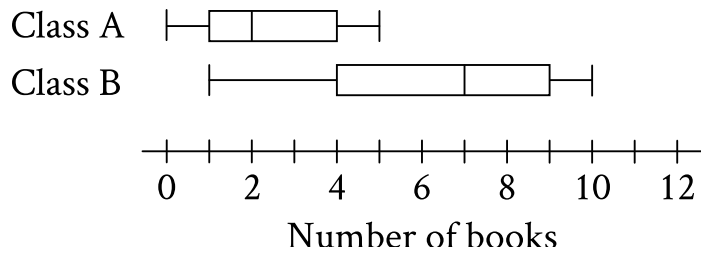


- The histogram has 4 bins.
- The frequency values of the 4 bins are as follows:
 - 30 to 39 points: More than halfway between 0 and 5
 - 40 to 49 points: About halfway between 5 and 10
 - 50 to 59 points: About halfway between 25 and 30
 - 60 to 69 points: About halfway between 10 and 15

The histogram summarizes data set A, which represents the number of points per player earned by 50 players of a game. A new player earns 18 points playing the game, and this number of points is added to data set A to create data set B with 51 values. Which of the following must be true?

- I. The median number of points per player for data set B is less than the median number of points per player for data set A.
 - II. The mean number of points per player for data set B is less than the mean number of points per player for data set A.
- A. I only
- B. II only
- C. I and II

D. Neither I nor II



- From left to right, the values of the vertical bars in the Class A box plot are as follows:
 - First vertical bar: 0
 - Second vertical bar: 1
 - Third vertical bar: 2
 - Fourth vertical bar: 4
 - Fifth vertical bar: 5
- From left to right, the values of the vertical bars in the Class B box plot are as follows:
 - First vertical bar: 1
 - Second vertical bar: 4
 - Third vertical bar: 7
 - Fourth vertical bar: 9
 - Fifth vertical bar: 10

The two box plots show the distribution of number of books read over the summer by the students in two different English classes. What is the positive difference between the ranges of number of books read over the summer for the two classes?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2

3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

For a certain computer game, individuals receive an integer score that ranges from 2 through 10. The table below shows the frequency distribution of the scores of the 9 players in group A and the 11 players in group B.

Score	Score Frequencies	
	Group A	Group B
2	1	0
3	1	0
4	2	0
5	1	4
6	3	2
7	0	0
8	0	2
9	1	1
10	0	2
Total	9	11

The median of the scores for group B is how much greater than the median of the scores for group A?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q4

GRID-IN

PROBLEM-SOLVING AND DATA ANALYSIS

Data set A consists of the heights of 75 objects and has a mean of 25 meters. Data set B consists of the heights of 50 objects and has a mean of 65 meters. Data set C consists of the heights of the 125 objects from data sets A and B. What is the mean, in meters, of data set C?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Percent of Residents Who Earned
a Bachelor's Degree or Higher

State	Percent of residents
State A	21.9%
State B	27.9%
State C	25.9%
State D	19.5%
State E	30.1%
State F	36.4%
State G	35.5%

A survey was given to residents of all 50 states asking if they had earned a bachelor's degree or higher. The results from 7 of the states are given in the table above. The median percent of residents who earned a bachelor's degree or higher for all 50 states was 26.95%. What is the difference between the median percent of residents who earned a bachelor's degree or higher for these 7 states and the median for all 50 states?

- A. 0.05%
- B. 0.95%
- C. 1.22%
- D. 7.45%

Each of the following frequency tables represents a data set. Which data set has the greatest mean?

A.

Value	Frequency
70	4
80	5
90	6
100	7

B.

Value	Frequency
70	6
80	6
90	6
100	6

C.

Value	Frequency
70	7
80	6
90	6
100	7

D.

Value	Frequency
70	8
80	5

90	5
100	8

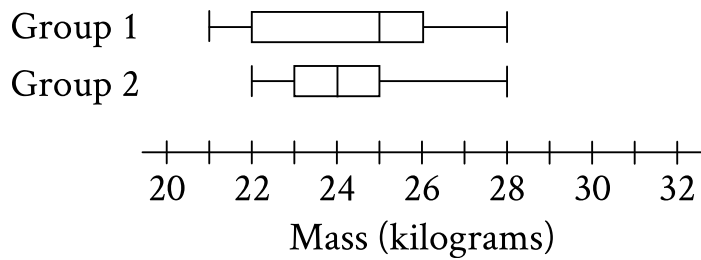
Value	Data set A frequency	Data set B frequency
30	2	9
34	4	7
38	5	5
42	7	4
46	9	2

Data set A and data set B each consist of 27 values. The table shows the frequencies of the values for each data set. Which of the following statements best compares the means of the two data sets?

- A. The mean of data set A is greater than the mean of data set B.
- B. The mean of data set A is less than the mean of data set B.
- C. The mean of data set A is equal to the mean of data set B.
- D. There is not enough information to compare the means of the data sets.

A scientist measured the lengths of 240 gray seals from Muskeget Island and 120 gray seals from Sable Island. The scientist determined that the mean length of the 240 gray seals from Muskeget Island was 88 inches and the mean length of the 120 gray seals from Sable Island was 94 inches. What was the mean length of all 360 gray seals the scientist measured for this study?

- A. 89
- B. 90
- C. 91
- D. 92

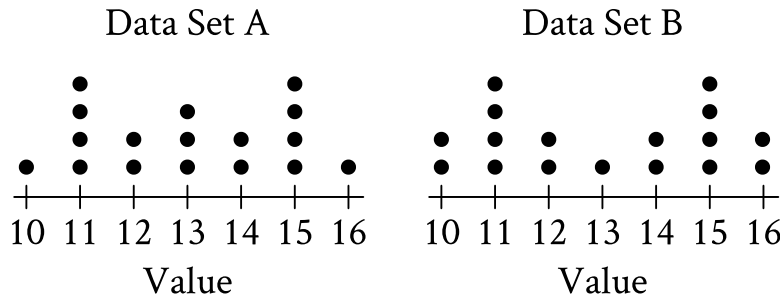


- From left to right the values of the vertical bars in the Group 1 box plot are as follows:
 - First vertical bar: 21
 - Second vertical bar: 22
 - Third vertical bar: 25
 - Fourth vertical bar: 26
 - Fifth vertical bar: 28
- From left to right the values of the vertical bars in the Group 2 box plot are as follows:
 - First vertical bar: 22
 - Second vertical bar: 23
 - Third vertical bar: 24
 - Fourth vertical bar: 25
 - Fifth vertical bar: 28

The box plots summarize the masses, in kilograms, of two groups of gazelles. Based on the box plots, which of the following statements must be true?

- A. The mean mass of group 1 is greater than the mean mass of group 2.
- B. The mean mass of group 1 is less than the mean mass of group 2.
- C. The median mass of group 1 is greater than the median mass of group 2.
- D. The median mass of group 1 is less than the median mass of group 2.

The dot plots represent the distributions of values in data sets A and B.



- For the dot plot titled Data Set A:
 - The number line ranges from 10 to 16 in increments of 1.
 - The data for the dot plot are as follows:
 - 10: 1 dot
 - 11: 4 dots
 - 12: 2 dots
 - 13: 3 dots
 - 14: 2 dots
 - 15: 4 dots
 - 16: 1 dot
- For the dot plot titled Data Set B:
 - The number line ranges from 10 to 16 in increments of 1.
 - The data for the dot plot are as follows:
 - 10: 2 dots
 - 11: 4 dots
 - 12: 2 dots
 - 13: 1 dot
 - 14: 2 dots
 - 15: 4 dots
 - 16: 2 dots

Which of the following statements must be true?

- I. The median of data set A is equal to the median of data set B.

II. The standard deviation of data set A is equal to the standard deviation of data set B.

- A. I only
- B. II only
- C. I and II
- D. Neither I nor II